## AD 2. AERODROMES

### VILK AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VILK - CHAUDHARY CHARAN SINGH INTERNATIONAL AIRPORT, LUCKNOW

### VILK AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

| | | | |
|---|---|---|---|
| 1 | Aerodrome reference point coordinates and its site | 264543N 0805300E<br>MAG BRG 78.26 DEG 842M from<br>physical extremity of RWY 09. | |
| 2 | Direction and distance of aerodrome reference point from the center of the city or town which the aerodrome serves | 210 DEG, 11KM from Lucknow Railway Station. | |
| 3 | Aerodrome elevation and reference temperature | 406 FT / 41.0 DEG C | |
| 4 | Magnetic variation, date of information and annual change | 0.25 DEG E (2010) /0.016 DEG E | |
| 5 | Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available) | Chief Airport Officer,<br>Lucknow International Airport Limited,<br>Ground Floor, Terminal-2,<br>Chaudhary Charan Singh International Airport, Lucknow,<br>Lucknow (U.P) PIN CODE-226009,, | |
| | | Telephone: | (91) 6358860132<br>(91) 6358860145 |
| | | Fax: | |
| | | AFS: | VILKYHYX |
| | | Email: | cao.lucknowairport@adani.com |
| 6 | Types of traffic permitted (IFR/VFR) | IFR/VFR | |
| 7 | Remarks | 1.https://lucknow.adaniairports.com/<br>2.Elevation in EGM 08<br>3. All military & non-scheduled flight operator shall file their slot request with AOCC Planning for clearance. All military/ non-scheduled movements shall be subjected to positive approval from Airport operations control centre (AOCC).<br>4.Request for International Movement must be forwarded at least 48hrs in advance.<br>5.Request for Domestic movement must be forwarded at least 24 hrs in advance.<br>Contact details of AOCC Planning are:<br>AOCC Mobile No +91 6358860132<br>Email: AOCC.LKO@ADANI.COM<br>Apron Control Mobile No +91 6358860145 | |

### VILK AD 2.3 OPERATIONAL HOURS

| | | |
|---|---|---|
| 1 | Aerodrome Operator | MON-FRI 0400-1300 UTC (0930-1830 IST), Sat 0400-0830 (0930-1400 IST)<br>Sun/Holiday: Nil |

| | | | |
|---|---|---|---|
| 2 | Custom and immigration | H24 for Sked Intl flights and see remark No. 12 for Non Sked Operations | |
| 3 | Health and sanitation | H24 | |
| 4 | AIS briefing office | H24 | |
| 5 | ATS reporting office (ARO) | H24 | |
| 6 | MET Briefing office | H24 | |
| 7 | Air Traffic Service | H24 | |
| 8 | Fuelling | H24 | |
| 9 | Handling | AIASL, Bird GSEC and Self Handling by Indigo | |
| 10 | Security | H24 | |
| 11 | De-icing | NIL | |
| 12 | Remarks | Customs and immigration facilities are provided to cover operations of scheduled INTL. Flights. The facilities can be arranged to cover any authorized non- sked operations on request with 24hrs PN. | |
| | | ATS approved hourly runway traffic handling capacity | A) From 0600 IST to 2200 IST daily (when approach control surveillance service is available):<br>1. For RWY 09:<br>Max number of arrival and departure: 17 (09 arrival and 08 departure)<br>Max number of arrivals only-09<br>Max number of departures only-20<br>2. For RWY 27:<br>Max number of arrival and departure: 17 (09 arrival and 08 departure)<br>Max number of arrivals only -20<br>Max number of departures only -12<br>B) From 2200 IST to 0600 IST daily (when procedural approach control is in progress.):<br>Max number of arrival and departure: 12 (06 arrival and 06 departure for both the runway)<br>Max number of arrivals only -06 (for both the runway)<br>Max number of departures only -10 (for both the runway) |

## VILK AD 2.4 HANDLING SERVICES AND FACILITIES

| | | |
|---|---|---|
| 1 | Cargo-handling facilities | 1. Facilities- International Cargo Terminal, Cool Room, Reefer Container Facility, Export utilization, DGR Storage Area,<br>2. Handling Equipment- Forklift (5 t), HPT, Slave Pallet, Pallet dolly, weighing scale, Bulk Trolleys<br>3. IT- Dedicated Cargo Management System (Galaxy). Air Cargo Community Portal. |
| 2 | Fuel and Oil types | JET A1 |

| | | |
|---|---|---|
| 3 | Fuelling facilities and capacity | Adani Aviation Fuel Service Limited<br>AFS 1- 200KLX4=800 KL<br>AFS-2- 73.5 KLX3=210.5 KL<br>Bowsers.16 KL X8 = 128KL, 25KLX2= 50 KL<br>Total Capacity inclusive of Bowsers – 1198.5 KL.<br>Duty Mobile - +918188068862 |
| 4 | De-icing facilities | NIL |
| 5 | Hangar space for visiting aircraft | NIL |
| 6 | Repair facilities for visiting aircraft | NIL |
| 7 | Remarks | Nil |

**VILK AD 2.5 PASSENGER FACILITIES**

| | | |
|---|---|---|
| 1 | Hotel(s) at or in the vicinity of aerodrome | In the City |
| 2 | Restaurant(s) at or in the vicinity of aerodrome | At AD and in the city |
| 3 | Transportation possibilities | Taxi, City Bus and Metro |
| 4 | Medical Facilities | First aid at AD. Hospital in the city |
| 5 | Bank and post office at or in the vicinity of aerodrome | Banks: 0930-1730 except sunday<br>Post office: 0930-1730 except sunday |
| 6 | Tourist office | At AD during ACFT Ops. |
| 7 | Remarks | Nil |

**VILK AD 2.6 RESCUE AND FIRE FIGHTING SERVICES**

| | | |
|---|---|---|
| 1 | Aerodrome category for fire fighting | Within ATS HR: CAT-7 CAT 8 is available on request by the Schedule operators with one-hour prior notice. |
| 2 | Rescue equipment | Available as per category. |
| 3 | Capability for removal of disabled aircraft | 1. The critical Aircraft for the Airport is A330<br>2. Primary responsibility for removal of disabled aircraft removal rests with the concerned airline.<br>3. The airlines may contact 'Aerodrome Coordinator for Disabled Aircraft Removal Operations' for coordinating with the agencies having Disabled Aircraft Recovery Kit.<br>4. Contact Details of Aerodrome Coordinator for Disabled Aircraft Removal Operations:<br>Aerodrome Coordinator: Head ARFF<br>Email: aocc.lko@adani.com<br>Contact No.: +91-6358860132, +91-6358860145 |
| 4 | Remarks | Nil |

**VILK AD 2.7 SEASONAL AVAILABILITY CLEARING**

| | | |
|---|---|---|
| 1 | Type(s) of clearing equipment | Manual and Mechanical Sweeper |
| 2 | Clearance priorities | High Priority-available H24 |
| 3 | Remarks | Nil |

**VILK AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA**

| | | |
|---|---|---|
| 1 | Designation, surface and strength of aprons | Refer Aircraft Parking/Docking Chart. |
| 2 | Designation, width, surface and strength of taxiways | Refer Aircraft Parking/Docking Chart. |
| 3 | Location and elevation of altimeter checkpoints | Location Not AVBL |
| 4 | Location of VOR checkpoints | Available on TWY A, B, C. and P1 |

| | | |
|---|---|---|
| 5 | Position of INS checkpoints | |
| 6 | Remarks | **1. Coordinates of RWY Holding Position:**<br>**TWY** **Coordinates**<br>A 264540.90N 0805313.35E<br>B 264540.75N 0805303.23E<br>C 264540.69N 0805258.80E<br>D 264540.75N 0805244.35E<br>E 264540.59N 0805242.77E<br>M 264534.32N 0805241.09E<br>P1 264540.60N 0805230.79E<br>P3 264540.65N 0805234.25E<br>P6 264541.43N 0805333.51E<br>**2. Coordinates of RWY & Taxi Intersection**<br>A 264537.87N 0805313.40E<br>B 264537.74N 0805303.28E<br>C 264537.68N 0805258.85E<br>D 264537.57N 0805250.74E<br>E/M 264537.44N 0805241.92E<br>P1 264537.32N 0805232.29E<br>P3 264537.34N 0805234.30E<br>P6 264538.11N 0805333.55E<br>3. Geoid Model: EGM08. |

## VILK AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

| | | |
|---|---|---|
| 1 | Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands | 1.TWY guidance signal at all the intersection with TWY and RWY and at all holding position Guidelines at Apron. Nose in guidance at aircraft stands.<br>2. Continuous aircraft stand lead-in lines available at MARS (Multiple Aircraft Ramp System) Centre stands for use of wide body aircraft and in single aircraft stands. Broken aircraft stand lead-in lines available at left & right MARS stand for narrow-bodied aircraft.<br>3. Stand identification signs are provided at all contact stands in Terminal-3 Aprons I. On remote stands, pole mounted stand identification signs have been provided in Apron I.<br>4. Docking of aircraft is done with the help of marshaller at remote parking stands and other contact stands where AVDGS is not provided or has become unserviceable. |
| 2 | Runway and taxiway markings and lights | RWY<br>Markings<br>Designation, THR, TDZ, centre line, aiming point,Runway side stripe.<br>Lights<br>RWY 27: THR,Wing Bars, TDZ, centre line, edge, end,CAT III Approach Lights<br>RWY 09: THR,Wing Bars, centre line, edge, end,SALS Approach Lights<br><br>TWY<br>Marking<br>Centre line, side strip, Runway holding position, Enhanced taxiway centre<br>line marking.<br>Lights<br>Centre line lights: TWY B,C,D, E.P1, P3 ,P6, part of parallel TWY P, L3 and TXL L & TXL L1.<br>Edge lights: i) TWY A,<br>ii) TWY B,C, D, E, P1, P3 ,P6, part of parallel TWY P & L3 have edge lights on curved portions only |

| | | |
|---|---|---|
| 3 | Stop bars (if any) | TWY A (105M from RWY CL),<br>TWY B (105M from RWY CL),<br>TWY C (105M from RWY CL),<br>TWY D (240M from RWY CL)<br>TWY E (105M from RWY CL)<br>TWY P1 (102M from RWY CL),<br>TWY P3 (105M from RWY CL)<br>TWY P6 (105M from RWY CL) |
| 4 | Remarks | 1.RWY guard lights are provided at TWY A, TWY B, TWY C, TWY D, TWY E, TWY P1 and TWY P3 and TWY P6.<br>2. An illuminated wind direction indicator in the form of truncated cone made of fabric of red and white colour in bands have been installed at north of RWY 09/27, 264543.9N 0805318.3E east of TWY A. |

**VILK AD 2.10 AERODROME OBSTACLES**

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation EGM 08 | Marking/LGT | Remarks |
| 27/TKOF<br>09/APCH | BUILDING | 264532.0N<br>0805209.7E | 450 FT | NIL | Building |
| 27/TKOF<br>09/APCH | BUILDING | 264533.6N<br>0805210.0E | 441 FT | NIL | Building |
| 27/TKOF<br>09/APCH | POLE | 264538.7N<br>0805220.9E | 419 FT | NIL | Camera Pole |
| 27/TKOF<br>09/APCH | POLE | 264530.8N<br>0805214.7E | 443 FT | NIL | Light Pole |
| 27/TKOF<br>09/APCH | POLE | 264531.6N<br>0805215.2E | 443 FT | NIL | Light Pole |
| 27/TKOF<br>09/APCH | TREE | 264540.2N<br>0805221.3E | 433 FT | NIL | Group of Tree |
| 27/TKOF<br>09/APCH | OTHER | 264530.5N<br>0805210.5E | 458 FT | LGTD | Cellphone Mast |
| 27/TKOF<br>09/APCH | OTHER | 264530.6N<br>0805210.6E | 463 FT | LGTD | Cellphone Mast |
| 27/TKOF<br>09/APCH | TREE | 264536.3N<br>0805210.0E | 455 FT | NIL | Group of Tree |
| 27/TKOF<br>09/APCH | BUILDING | 264532.4N<br>0805210.9E | 444 FT | NIL | Building |
| 27/TKOF<br>09/APCH | TREE | 264533.8N<br>0805215.8E | 435 FT | NIL | Group of Tree |
| 27/TKOF<br>09/APCH | OTHER | 264537.1N<br>0805219.0E | 423 FT | NIL | Mobile Road Traffic (Road Elev.123.8m+5m Mobile Traffic Ht.) |
| 27/TKOF<br>09/APCH | TREE | 264535.1N<br>0805211.0E | 451 FT | NIL | Group of Tree |
| 27/TKOF<br>09/APCH | OTHER | 264541.1N<br>0805221.5E | 423 FT | NIL | Mobile Road Traffic (Road Elev.123.8m+5m Mobile Traffic Ht.) |
| 27/TKOF<br>09/APCH | OTHER | 264530.6N<br>0805210.5E | 444 FT | LGTD | Cellphone Mast |
| 27/TKOF<br>09/APCH | POLE | 264530.8N<br>0805214.7E | 442 FT | NIL | Light Pole |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation EGM 08 | Marking/LGT | Remarks |
| In circling area and at AD | POLE | 264531.6N 0805215.2E | 442 FT | NIL | Light Pole |
| 27/TKOF 09/APCH | OTHER | 264539.4N 0805220.6E | 421 FT | NIL | Mobile Road Traffic (Road Elev.123.3m+5m Mobile Traffic Ht.) |
| 27/TKOF 09/APCH | TREE | 264540.0N 0805221.2E | 436 FT | NIL | Group of Tree |
| 27/TKOF 09/APCH | BUILDING | 264533.8N 0805205.7E | 447 FT | NIL | Sports Authority of India |
| 27/TKOF 09/APCH | OTHER | 264542.4N 0805223.3E | 419 FT | NIL | Mobile Road Traffic (Road Elev. 122.6m+ 5m Mobile Traffic Ht.) |
| 27/TKOF 09/APCH | TREE | 264532.0N 0805214.8E | 435 FT | NIL | Tree |
| 27/TKOF 09/APCH | TREE | 264536.6N 0805217.9E | 438 FT | NIL | Tree |
| 27/APCH 09/TKOF | ANTENNA | 264536.9N 0805421.5E | 424 FT | LGTD | Antenna On FM Hut |
| 27/APCH 09/TKOF | TREE | 264540.7N 0805423.3E | 430 FT | NIL | Group of Tree |
| 27/APCH 09/TKOF | POLE | 264534.5N 0805418.3E | 417 FT | NIL | Light With Solar Panel |
| 27/APCH 09/TKOF | TREE | 264534.3N 0805420.4E | 416 FT | NIL | Group of Tree |
| 27/APCH 09/TKOF | TREE | 264542.6N 0805421.9E | 445 FT | NIL | Group of Tree |
| 27/APCH 09/TKOF | TREE | 264540.8N 0805424.1E | 426 FT | NIL | Group of Tree |
| 27/APCH 09/TKOF | TREE | 264540.8N 0805431.4E | 437 FT | NIL | Group of Tree |
| In circling area and at AD | BUILDING | 264530.2N 0805213.4E | 441 FT | NIL | Building |
| In circling area and at AD | TREE | 264544.2N 0805214.0E | 484 FT | NIL | Group of Tree |
| In circling area and at AD | TREE | 264544.5N 0805221.9E | 496 FT | NIL | Group of Tree |
| In circling area and at AD | TREE | 264544.2N 0805217.8E | 491 FT | NIL | Group of Tree |
| In circling area and at AD | OTHER | 264542.6N 0805222.3E | 420 FT | NIL | Mobile Road Traffic (Road Elev.123.0m+5m Mobile Traffic Ht.) |
| In circling area and at AD | OTHER | 264537.3N 0805229.7E | 405 FT | NIL | Approach Light |
| In circling area and at AD | BUILDING | 264529.9N 0805212.2E | 441 FT | NIL | Building Top |
| In circling area and at AD | POLE | 264543.5N 0805222.8E | 440 FT | NIL | Light Pole |
| In circling area and at AD | OTHER | 264536.3N 0805249.4E | 406 FT | NIL | PAPI |
| In circling area and at AD | OTHER | 264534.4N 0805242.9E | 406 FT | NIL | Electrical Pillar Box |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation EGM 08 | Marking/LGT | Remarks |
| In circling area and at AD | OTHER | 264533.2N 0805241.5E | 407 FT | NIL | Isolation Bay Light |
| In circling area and at AD | TREE | 264544.9N 0805222.5E | 506 FT | NIL | Group of Tree |
| In circling area and at AD | TREE | 264543.9N 0805223.9E | 442 FT | NIL | Group of Tree |
| In circling area and at AD | POLE | 264529.9N 0805214.1E | 441 FT | NIL | Light Pole |
| In circling area and at AD | POLE | 264544.3N 0805223.3E | 441 FT | NIL | Light Pole |
| In circling area and at AD | POLE | 264542.6N 0805222.2E | 441 FT | NIL | Light Pole |
| In circling area and at AD | NAVAID | 264542.7N 0805236.0E | 431 FT | LGTD | SMR Antenna |
| In circling area and at AD | TREE | 264544.2N 0805216.5E | 471 FT | NIL | Tree |
| In circling area and at AD | TREE | 264544.2N 0805214.2E | 405 FT | NIL | Tree |
| In circling area and at AD | TREE | 264542.8N 0805236.2E | 421 FT | NIL | Tree |
| In circling area and at AD | NAVAID | 264542.8N 0805402.3E | 426 FT | LGTD | G.P. Monitor Antenna |
| In circling area and at AD | NAVAID | 264545.7N 0805417.5E | 438 FT | LGTD | SMR Antenna |
| In circling area and at AD | TREE | 264531.1N 0805355.9E | 466 FT | NIL | Group of Tree |
| In circling area and at AD | TREE | 264545.8N 0805409.0E | 445 FT | NIL | Group of Tree |
| In circling area and at AD | TREE | 264545.4N 0805422.4E | 443 FT | NIL | Group of Tree |
| In circling area and at AD | NAVAID | 264535.7N 0805358.6E | 420 FT | NIL | Met Antenna |
| In circling area and at AD | OTHER | 264537.2N 0805357.8E | 404 FT | NIL | PAPI |
| In circling area and at AD | NAVAID | 264542.8N 0805359.3E | 460 FT | LGTD | GP Antenna(DME Top-130.1m) |
| In circling area and at AD | POLE | 264542.4N 0805401.4E | 404 FT | LGTD | Pole |
| In circling area and at AD | TREE | 264545.9N 0805405.9E | 448 FT | NIL | Group of Tree |
| In circling area and at AD | NAVAID | 264535.6N 0805400.9E | 410 FT | NIL | Met Antenna |
| In circling area and at AD | TREE | 264531.6N 0805320.9E | 439 FT | NIL | Tree |
| In circling area and at AD | NAVAID | 264535.0N 0805400.9E | 416 FT | NIL | Met Antenna |
| In circling area and at AD | NAVAID | 264535.1N 0805359.8E | 415 FT | NIL | Met Antenna |
| In circling area and at AD | TREE | 264531.8N 0805355.6E | 467 FT | NIL | Group of Tree |
| In circling area and at AD | TREE | 264545.9N 0805402.4E | 452 FT | NIL | Group of Tree |

| In Approach/Take-off/Circling Area and at AD | | | | | |
|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** |
| RWY/Area affected | Obstacle type | Coordinates | Elevation EGM 08 | Marking/LGT | Remarks |
| In circling area and at AD | BUILDING | 264542.8N 0805359.1E | 419 FT | LGTD | GP Hut |
| In circling area and at AD | POLE | 264543.1N 0805359.3E | 424 FT | NIL | Camera Pole |
| In circling area and at AD | OTHER | 264531.5N 0805340.5E | 450 FT | NIL | Arrestor Rod |
| In circling area and at AD | OTHER | 264543.9N 0805318.3E | 429 FT | LGTD | WDI |
| In circling area and at AD | POLE | 264542.9N 0805311.2E | 420 FT | NIL | Pole |
| In circling area and at AD | OTHER | 264533.8N 0805246.4E | 421 FT | NIL | Light Arrestor |
| In circling area and at AD | OTHER | 264533.9N 0805245.9E | 414 FT | NIL | Light Arrestor |
| In circling area and at AD | NAVAID | 264534.2N 0805318.5E | 408 FT | NIL | Met Antenna |
| In circling area and at AD | NAVAID | 264534.2N 0805317.4E | 408 FT | NIL | Met Antenna |
| In circling area and at AD | NAVAID | 264534.3N 0805322.9E | 406 FT | NIL | Met Antenna |
| In circling area and at AD | BUILDING | 264532.8N 0805340.0E | 419 FT | NIL | Security Hut |
| In circling area and at AD | OTHER | 264655.0N 0805313.9E | 557 FT | LGTD | BSNL Tower |
| In circling area and at AD | TOWER | 264437.6N 0805135.4E | 599 FT | LGTD | Tower |
| In circling area and at AD | OTHER | 264439.7N 0805104.2E | 557 FT | NIL | Pylon Mast |
| In circling area and at AD | OTHER | 264434.0N 0805104.4E | 580 FT | NIL | Pylon Mast |
| In circling area and at AD | TOWER | 264432.1N 0805147.8E | 589 FT | LGTD | Tower |
| In circling area and at AD | TOWER | 264426.9N 0805149.0E | 610 FT | LGTD | Tower |
| In circling area and at AD | TOWER | 265224.3N 0805156.9E | 982 FT | LGTD | TV Tower |

## VILK AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

| | | |
|---|---|---|
| 1 | Name of the associated meteorological office | Lucknow |
| 2 | Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours | H24 |
| 3 | Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts | Lucknow<br>9,24HR |
| 4 | Availability of the trend forecast for the aerodrome and interval of issuance | Trend forecast BTN 0000 to 1600UTC half hourly. BTN 1630 to 2330UTC forecast hourly without trend. |
| 5 | Information on how briefing and/or consultation is provided | Provided |
| 6 | Types of flight documentation supplied and language(s) used in flight documentation | Charts and Tabular form. English WAFC Product. |

| | | |
|---|---|---|
| 7 | Charts and other information displayed or available for briefing or consultation | S, U85, U70, U50, U30, U20 |
| 8 | Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images; | Three new AWOS (Automatic Weather Observing System) installed at RWY 27 and commissioned for TDZ, MID, and END RVR of RWY 27. |
| 9 | The air traffic services unit(s) provided with meteorological information | Lucknow ATC and ACS |
| 10 | Additional information, e.g. concerning any limitation of service. | Operational hour of Met briefing: 2330-1630 Hrs UTC. |

## VILK AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

| Designations | TRUE Bearings | Dimensions of RWY (M) | Strength of pavement (PCR) and associated data) and surface of runway and associated stopways | Geographical coordinates for threshold and runway end |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| 09 | 89.14 DEG | 2744 x 45 M | | THR:<br>264537.36N<br>0805236.13E<br>RWY END:<br>264538.62N<br>0805409.75E |
| 27 | 269.14 DEG | 2744 x 45 M | | THR:<br>264538.62N<br>0805409.75E<br>RWY END:<br>264537.29N<br>0805230.45E |

| THR elevation and highest elevation of TDZ of precision APP RWY | Slope of runway and associated stopway | Dimensions of stopway (M) | Dimensions of clearway (M) | Dimensions of strips (M) |
|---|---|---|---|---|
| 6 | 7 | 8 | 9 | 10 |
| THR: 405.0FT<br>TDZ: 406.0FT | 0.06% | NIL | NIL | 2864 x 280 M |
| THR: 400.0FT<br>TDZ: 404.0FT | 0.06% | NIL | NIL | 2864 x 280 M |

| Dimensions of runway end safety areas | Location and description of arresting system (if any) | Existence of an obstacle-free zone | Remarks. |
|---|---|---|---|
| 11 | 12 | 13 | 14 |
| 115M x 90M | | AVBL | 1. APCH Slope - 1:50<br>2. Surface: Asphalt<br>3.Geoid Model: EGM 08<br>4.PCR-1181/F/C/W/T - ASPHALT |
| 182M x 90M | | AVBL | 1. APCH Slope - 1:50<br>2. Surface: Asphalt<br>3. Geoid Model: EGM 08<br>4.PCR-1181/F/C/W/T - ASPHALT |

## VILK AD 2.13 DECLARED DISTANCES

| RWY Designator | Take-off run available TORA (M) | Take-off distance available TODA (M) | Accelerate distance available ASDA (M) | Landing distance available LDA (M) | Remarks (including runway entry or start point where alternative reduced declared distances have been declared) |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| 09 | 2744 | 2744 | 2744 | 2587 | RWY09 THR displaced by 157M.<br>TWY A: 1557 M (TORA, TODA, ASDA)<br>TWY B: 1837 M (TORA, TODA, ASDA)<br>TWY C: 1959 M (TORA, TODA, ASDA)<br>TWY D: 2183 M (TORA, TODA, ASDA)<br>TWY E: 2427 M (TORA, TODA, ASDA)<br>TWY M: 2427 M (TORA, TODA, ASDA)<br>TWY P3: 2637 M (TORA, TODA, ASDA) |
| 27 | 2744 | 2744 | 2744 | 2744 | TWY A: 1187 M (TORA, TODA, ASDA)<br>TWY B: 907 M (TORA, TODA, ASDA)<br>TWY C: 784 M (TORA, TODA, ASDA)<br>TWY D: 560 M (TORA, TODA, ASDA)<br>TWY E: 317 M (TORA, TODA, ASDA)<br>TWY M: 317 M (TORA, TODA, ASDA)<br>TWY P3: 107 M (TORA, TODA, ASDA)<br>Intermediate turn pad: 2214 M |

## VILK AD 2.14 APPROACH AND RUNWAY LIGHTING

| Runway Designator | Type, length and intensity of approach lighting system | Runway threshold lights, colour and wing bars | Type of visual slope indicator system | Length of runway touchdown zone lights |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| **09** | SALS<br>420 M<br>LIH | Green<br>AVBL | PAPI<br>RIGHT/3.00 DEG<br>MEHT<br>(58.09FT) | |
| **27** | CAT III<br>900 M<br>LIH | Green<br>AVBL | PAPI<br>LEFT/3.00 DEG<br>MEHT<br>(54.43FT) | 900 M |

| Length, spacing, colour and intensity of runway centre line lights | Length, spacing, colour and intensity of runway edge lights | Colour of runway end lights and wing bars | Length and colour of stopway lights | Remarks |
|---|---|---|---|---|
| 6 | 7 | 8 | 9 | 10 |
| 2744 M<br>15 M<br>LIH | 2744 M<br>60 M<br>LIH | Red | NIL | 1. Runway centreline lights variable white from threshold to the point 900m before the runway end; alternate red and variable white from 900m to 300m from the runway end, and red from 300m to the runway end.<br>2. RWY Edge lights from physical beginning to displaced THR red. Variable white from THR up to 600m before runway end; Yellow between 600m and runway end. |
| 2744 M<br>15 M<br>LIH | 2744 M<br>60 M<br>LIH | Red | NIL | 1. Runway centreline lights variable white from threshold to the point 900m before the runway end; alternate red and variable white from 900m to 300m from the runway end, and red from 300m to the runway end.<br>2. Runway edge lights variable white from physical beginning up to 600m before runway end; Yellow between 600m and runway end |

## VILK AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

| | | | |
|---|---|---|---|
| 1 | Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any) | ABN | At tower building, FLG W&G EV 3SEC, H24 |
| | | IBN | NIL |

| | | | |
|---|---|---|---|
| 2 | Location and lighting (if any) of anemometer/ landing direction indicator; | LDI | 56M NNE of ARP,Lighted |
| | | Anemometer | Available |
| 3 | Taxiway edge and taxiway centre line lights; | Edge | ALL TWY |
| | | Centre Line | TWY B, C, D, E, P1, P3 , P6,P ,L1 AND L3 |
| 4 | Secondary power supply including switch-over time; | Secondary power supply to all lighting at AD Switch-over time: CAT I/ CAT II/CAT III: 1 Second<br>Online UPS available for all circuits. | |
| 5 | Remarks | Aircraft stand manoeuvring guidance light provided from Aircraft stand 101L up-to Aircraft stand 14 and stand 212 to stand 216. | |

**VILK AD 2.16 HELICOPTER LANDING AREA**

| | | |
|---|---|---|
| 1 | Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area | Not established |
| 2 | TLOF and/or FATO area elevation: | Not established |
| 3 | TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking; | Not established |
| 4 | True bearings of FATO; | Not established |
| 5 | Declared distances available | Not established |
| 6 | Approach and FATO lighting; | Not established |
| 7 | Remarks | Not established |

**VILK AD 2.17 AIR TRAFFIC SERVICE AIRSPACE**

| | | |
|---|---|---|
| 1 | Airspace designation, geographical coordinates and lateral limits | CTR: Circular area centered on 264531N 0805340E within a 25NM radius. |
| 2 | Vertical limits | FL 105 |
| 3 | Airspace classification | D |
| 4 | Call sign and language(s) of the air traffic services unit providing service; | Lucknow Tower / Lucknow Approach, English |
| 5 | Transition altitude | 4000 FT |
| 6 | Hours of applicability | HO |
| 7 | Remarks | Nil |

## VILK AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

| Service Designation | Call sign | Channel(s) | SATVOICE Number(s), if available |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| APP | Lucknow Approach | 124.800 MHZ | |
| APP | Lucknow Approach | 125.800 MHZ | |
| TWR | Lucknow Tower | 118.600 MHZ | |
| TWR | Lucknow Tower | 122.500 MHZ | |
| ATIS | Lucknow | 126.800 MHZ | |
| ACC | Lucknow Control | 120.200 MHZ | |
| ACC | Lucknow Control | 120.450 MHZ | |
| ALRS | Emergency Frequency | 121.500 MHZ | |
| SMC | Lucknow Ground | 121.625 MHZ | |

| Logon address, as appropriate | Hours of operation | Remarks |
|---|---|---|
| 5 | 6 | 7 |
| | H24 | SDBY FREQ |
| | H24 | MAIN FREQ |
| | H24 | MAIN FREQ |
| | H24 | SDBY FREQ |
| | H24 | BTN 0130 - 1630 UTC half Hourly 1630 - 0130 UTC Hourly |
| | H24 | BACKUP FREQ. |
| | H24 | NIL |
| | H24 | NIL |
| | H24 | MAIN |

## VILK AD 2.19 RADIO NAVIGATION AND LANDING AIDS

| Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid | Identification | Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate | Hours of operation, as appropriate; |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| LOC 27<br>CAT IIIB | ILUC | 109.900 MHz | H24 |
| GP 27 | ILUC | 333.800 MHz | H24 |
| DME ILS 27 | ILUC | CH36X | H24 |
| DVOR/DME | LKN | 117.400 MHz<br>CH121X | H24 |

| Geographical coordinates of the position of the transmitting antenna | Elevation of transmitting antenna of DME/ elevation of GBAS reference point | Service volume radius from the GBAS reference point | Remarks |
|---|---|---|---|
| 5 | 6 | 7 | 8 |
| 264537.2N<br>0805221.2E | | | ILS CAT-IIIB |
| 264542.8N<br>0805359.3E | | | 3 DEG |
| 264542.8N<br>0805359.3E | 427 FT | | Collocated with GP27<br>Geoid Model: EGM 08 |
| 264531.0N<br>0805340.0E | 440 FT | | |

## VILK AD 2.20 LOCAL AERODROME REGULATIONS

**I. PUSHBACK PROCEDURES:**

Pushback and taxi Procedure for stand 101L, 101, 101R, 102L, 102, 102R,103L, 103, 103R, 104L, 104, 104R, 105L, 105R, 106L, 7 to 14 and 212to 216.

**i. Stand No. 103L**

a) Pushback and startup may be given facing west abm stand 103. Taxi out via TWY P1/P3

b) Pushback not permitted on stand 103R, 104L, 104 and 104R.

c) Docking not permitted on stand 103, 103R, 104L, 104 and 104R.

d) Pushback and startup may be given facing east on tug release point (TRP) 03. Taxi out via twy E/D/C/B.

**ii. Stand No. 103**

a) Pushback and startup may be given facing west abm stand 103R. Taxi out via twy P1/P3. TWY D/E not available.

b) Pushback not permitted on stand 104L, 104, 104R, 105L and 105R.

c) Docking not permitted on stand 104L, 104 and 104R.

d) Pushback and startup may be given facing east on TRP 03. Taxi out via TWY E/D/C/B.

**iii. Stand No. 103R.**

a) Pushback and startup may be given facing west abm bay 104L. Taxi out via TWY P1/P3. TWY D/E not available.

b) Pushback not permitted on stand 103L, 104L, 104 and 104R.

c) Docking not permitted on stand 104L and 104.

d) Pushback and startup may be given facing east on TRP 03. Taxi out via TWY E/D/C/B.

**iv. Stand No. 104L**

a) Pushback and Startup may be given facing East abeam stand 105L or pull abeam stand 106L to accommodate more pushbacks from stand 103L, 103R and Taxi out via TWY B/C/D/E. or facing West abeam bay 105L Taxi out via TWY D/E

b) Pushback not permitted on Stand 104R, 105L and 105R.

c) Docking not permitted on Stand 105L or stand 105R.

**v. Stand No. 104**

a) Pushback and Startup may be given facing East abeam stand 106L and Taxi out via TWY B/C or facing West abeam bay 105L Taxi out via TWY D/E.

b) Pushback not permitted on Stand 103L, 103 103R, 105L and 105R, 106L.

c) Docking not permitted on Stand 105L and stand 105R.

**vi. Stand No. 104R**

a) Pushback and Startup may be given facing East abeam stand 105L and Taxi out via TWY B/C or facing West abeam bay 105L Taxi out via TWY D/E

b) Pushback not permitted on Stand 103L, 103 103R, 104L, 105L.

c) Docking not permitted on Stand 105L.

**vii. Stand No. 105L**

a) Pushback and Startup may be given facing East abeam stand 105R and Taxi out via TWY B/C or facing West abeam bay 105R Taxi out via TWY D/E.

b) Pushback not permitted on 103L, 103 103R, 104L, 104, 104R and 105R.

c) Docking is not permitted on Stand 105L or 105R

**viii. Stand No. 105R**

a) Pushback and Startup may be given facing East abeam stand 105L and Taxi out via TWY B/C or facing West abeam bay 105L Taxi out via TWY D/E.

b) Pushback not permitted on Stand 103L, 103 103R, 104L, 104, 104R, 105L.

c) Docking not permitted on Stand 105L.

**ix. Stand No. 106L**

a) Pushback and Startup may be given facing East abeam stand 105R and Taxi out via TWY B/C or facing West abeam bay 105R Taxi out via TWY D/E.

b) Pushback not permitted on Stand 103L, 103 103R, 104L, 104, 104R, 105L,105R.

c) Docking not permitted on Stand 105L.

**x. Stand No. 12**

a) Pushback and Startup may be given facing East abeam stand 12 and Taxi out via TWY B or facing West abeam bay 09 Taxi out via TWY D/E.

b) Pushback not permitted on Stand 10, 11, 13, & 14.

c) Docking not permitted on Stand 11, 13, & 14.

**xi. Stand No. 07**

a) Pushback may be given facing West or East depending upon the traffic flow on taxiways and RWY in use.

b) Pushback and Startup facing east abeam bay no.07.

c) Taxi out via taxiway C or B.

d) Pushback and Startup may be given facing west abeam bay no.07.

e) Taxi out via taxiway D/E.

**xii. Stand No. 08**

a) Pushback may be given facing West or East depending upon the traffic flow on taxiways and RWY in use.

b) Pushback and Startup facing east abeam bay no.08.

c) Taxi out via taxiway C or B.

d) Pushback and Startup may be given facing west abeam bay no.08.

e) Taxi out via taxiway D/E.

**xiii. Stand No. 09**

a) Pushback may be given facing West or East depending upon the traffic flow on taxiway 'B' and 'C' and RWY in use.

b) Pushback and Startup facing east abeam bay no. 09.

c) Taxi out via taxiway C or B.

d) Pushback and Startup may be given facing west abeam bay no.09.

e) Taxi out via taxiway D/E.

**xiv. Stand No. 10**

a) Pushback may be given facing West or East depending upon the traffic flow on taxiway 'B' and 'C' and RWY in use.

b) Pushback and Startup facing east abeam bay no.10.

c) Taxi out via taxiway C or B.

d) Taxiway C is blocked when aircraft pushes back abeam bay no.10.

e) No arrival/departure shall taxi in/out via taxiway C.

f) Pushback and Startup may be given facing west abeam bay no.10.

g) Taxi out via taxiway D/E.

h) Taxi way C is blocked when aircraft pushes back abeam bay no.10.

i) No arrival/departure shall taxi in/out via taxiway C.

**xv. Stand No. 11**

a) Pushback may be given facing West or East depending upon the traffic flow on taxiway 'B' and 'C' and RWY in use.

b) Pushback and Startup facing east abeam bay no.11.

c) Taxi out via taxiway B only.

d) Taxiway C is blocked when aircraft pushes back abeam bay no.11.

e) No arrival/departure shall taxi in/out via Taxiway C.

f) For taxing out via C the aircraft may be advise to continue the Pushback abeam bay No 10.

g) Pushback and Startup may be given facing west abeam bay no.11.

h) Taxi out via taxiway D/E.

i) Taxiway C is blocked when aircraft pushes back abeam bay no.11.

j) No arrival/departure shall taxi in/out via taxiway C.

**xvi. Stand No. 13 (Code D/E Aircraft)**

A) Pushback may be given facing west or east depending upon the traffic flow on taxiway B and C

B) Pushback and startup facing east abeam bay no.12.

C) Taxi out via taxiway B.

D) Taxiway C is blocked when the aircraft pushes back abeam bay no. 12.

E) No arrival/departure shall taxi in/out via taxiway C.

**xvii. Stand No. 13 (Code C Aircraft)**

a) Pushback may be given facing West or East depending upon the traffic flow on taxiway 'B' and 'C.'

b) Pushback and Startup facing east abeam bay no. 12.

c) Taxi out via taxiway B.

d) Taxiway C will be blocked when aircraft pushed back abeam bay no.12.

e) No arrival/departure shall taxi in/out via Taxiway C.

f) Pushback and Startup may be given facing west abeam bay no.13.

g) Taxi out via taxiway C/D/E.

h) Taxiway B is blocked when aircraft pushes back abeam bay no.13.

**xviii. Stand No. 14 (Code C / D/E Aircraft)**

a) Pushback and Startup shall be facing only East abeam bay no. 12.

b) Taxi out via taxiway B.

c) Taxiway C is blocked when aircraft pushes back abeam bay no.12.

d) No arrival/departure shall taxi in/out via Taxiway C.

**xix. Stand 212**

a) Pushback and Startup shall be given facing East on TXL L1 and pull abeam stand 213 for Startup.

b) Pushback not permitted on Stand 213, 214 and 215.

c) Docking not permitted on Stand 213 and 214.

**xx. Stand 213**

a) Pushback and Startup shall be given facing East on TXL L1.

b) Pushback not permitted on Stand 212, 214 and 215.

c) Docking not permitted on Stand 212, 214 and 215.

**xxi. Stand 214**

a) Pushback and Startup shall be given facing East on TXL L1.

b) Pushback not permitted on Stand 212, 213, 215 and 216.

c) Docking not permitted on Stand 212, 213 and 215.

**xxii. Stand 215**

a) Pushback and Startup shall be given facing East on TXL L1.

b) Pushback not permitted on Stand 212, 213, 214 and 216.

c) Docking not permitted on Stand 212, 213, 214, and 216.

**xxiii. Stand 216**

a) Pushback shall be given facing East on TXL L1.

b) Pushback not permitted on Stand 215, 214, 213 and 212.

c) Docking not permitted on Stand 215, 214, 213 and 212.

**xxiv. Stand 101L**

a) Pushback and Startup may be given facing west on TXL L. Taxi out via TWY P1. TWY P3 and TRP 3 will be blocked.

b) Pushback not permitted on Stand 101R, 102 and 102L.

c) Docking not permitted on Stand 101R.

d) Pushback and Startup may be given facing East on TXL L. Taxi out via TWY E/ D/ C/ B. TWY P3 will be blocked.

e) Pushback not permitted on Stand 101R, 102, 102L and 103.

f) Docking not permitted on Stand 101 and 101R

**xxv. Stand 101**

a) Pushback and Startup may be given facing west on TXL L. Taxi out via TWY P1. TWY P3 and TRP 3 will be blocked.

b) Pushback not permitted on Stand 102L, 102 and 102R.

c) Docking not permitted on Stand 101L, 101R, 102L and 102.

d) Pushback and Startup may be given facing East on TXL L. Taxi out via TWY E/ D/ C/ B. TWY P3 and TRP 3 will be blocked.

e) Pushback not permitted on Stand 102L, 102, 102R, 103L and 103.

f) Docking not permitted on Stand 101L and 101R..

**xxvi. Stand 101R**

a) Pushback and Startup may be given facing west on TXL L. Taxi out via TWY P1/ P3. TWY P3 and TRP 3 will be blocked.

b) Pushback not permitted on Stand 101L, 102L, 102 and 102R.

c) Docking not permitted on Stand 101, 102L and 102.

d) Pushback and Startup may be given facing East on TXL L. Taxi out via TWY E/ D/ C/ B. TWY P3 and TRP 3 will be blocked.

e) Pushback not permitted on Stand 101L, 102L, 102, 102R, 103L and103.

f) Docking not permitted on Stand 101, 101L and 102L.

**xxvii. Stand 102L**

a) Pushback and Startup may be given facing west on TXL L. Taxi out via TWY P1/ P3. TRP 3 will be blocked.

b) Pushback not permitted on Stand 101L, 101, 101R, and 102R.

c) Docking not permitted on Stand 102 and 102R.

d) Pushback and Startup may be given facing East on TXL L. Taxi out via TWY E/ D/ C/ B. TRP 3 will be blocked.

e) Pushback not permitted on Stand 101L, 101, 101R, 102R, 103L and 103.

f) Docking not permitted on Stand 101, 101R and 102.

**xxviii. Stand 102**

a) Pushback and Startup may be given facing west on TXL L, pull abeam stand 102L. Taxi out via TWY P1/ P3. TRP 3 will be blocked.

b) Pushback not permitted on Stand 101, 102R, 103L and 103.

c) Docking not permitted on Stand 102R.

d) Pushback and Startup may be given facing East on TXL L. Taxi out via TWY E/ D/ C/ B.

e) Pushback not permitted on Stand 101L, 101, 101R, 103L and 103.

f) Docking not permitted on Stand 101L, 101 and, 101R , 102L and 102R

**xxix. Stand 102R**

a) Pushback and Startup may be given facing west on TXL L, pullabeam stand 102L. Taxi out via TWY P1/ P3. TRP 3 will be blocked.

b) Pushback not permitted on Stand 101, 101R, 102L and 103L.

c) Docking not permitted on Stand 102, 102L and 102R.

d) Pushback and Startup may be given facing East on TXL L. Taxi out via TWY E/ D/ C/ B. TRP 3 will be blocked.

e) Pushback not permitted on Stand 101, 101R, 102L, 103L and 103.

f) Docking not permitted on Stand 102 and 102L.

**II. PRECAUTIONS TO BE TAKEN WHILE GIVING PUSHBACK:**

• Stand 101L, 101, 101R, 102L, 102, 102R, 103L, 103, 103R, 104L, 104, 104R, 105L, 105R, 106L, 212, 213, 214, 215 and 216 have been provided with AVDGS. A-VDGS will be operational,24X7.

• When a Code E aircraft is parked on stand 101, code C aircraft cannot be parked on Stand 101L and 101R.

• When a Code E aircraft is parked on stand 102, code C aircraft cannot be parked on Stand 102L and 102R.

• When a Code E aircraft is parked on stand 103, code C aircraft cannotbe parked on Stand 103L and 103R.

• When a Code E aircraft is parked on stand 104, code C aircraft cannot be parked on Stand 104L and 104R.

• Stand 12 is interdependent with Stand 13, if Code D/E parked on stand 12 only Code C can be parked on Stand 13 and vice versa.

**III.Radio communication failure (RCF) procedure.**

**1. INTRODUCTION:**

Radio communication failure (RCF) procedures are prescribed in Para 15.3 of ICAO PANS-ATM DOC 4444. Based on these provisions, following radio communication failure procedures are established, to standardize the actions to be taken by the pilot of arriving and departing aircraft at Lucknow airport.

**2. GENERAL**

2.1 All Transponder equipped aircraft experiencing Radio Communication Failure shall set transponder to Mode A/C code 7600 as soon as practicable.

2.2 ADS-B-equipped aircraft experiencing radio communication failure may transmit the appropriate ADS-B emergency and/ or urgency mode.

2.3 If two-way communication is lost with an aircraft, the controller should determine whether or not the aircraft's receiver is functioning by instructing the aircraft on the channel so far used to acknowledge by making a specified manoeuvre and by observing the aircraft's track, or by instructing the aircraft to operate IDENT or to make SSR code
and/or ADSB transmission changes.

2.4 If the action prescribed in 2.3 is unsuccessful, it shall be repeated on any other available channel on which it is believed that the aircraft might be listening.

2.5 In both the cases covered by 2.3 &2.4, any manoeuvring instructions shall be such that the aircraft would not enter into restricted airspace after having complied with the instructions received.

2.6 Where it has been established by the action in 2.3 that the aircraft's radio receiver is functioning, continued control can be effected using SSR code/ADS-B transmission changes or IDENT transmissions to obtain acknowledgement of clearances issued to the aircraft.

2.7 If the aircraft experiencing the communication failure is not identified, separation shall be applied in accordance with para 8.7.3 of PANS-ATM, between identified aircraft and all unidentified aircraft.

(Note: This Requirement of Setting transponder to Mode A/C code 7600 in no way imposes any restriction on the pilot's decision to set transponder to Mode A/C code 7500 or 7700, whenever required)

**3. ASSIGNED RUNWAY AND ITS AVAILABILITY FOR RCF AIRCRAFT:**

3.1 In case of arriving aircraft, when Runway for landing has already been advised to the aircraft by ATC, such runway shall be considered as assigned runway. If radio communication failure occurs after the receipt of final /interception turn or approach clearance, the aircraft may continue its approach to land on such Runway.

3.2 In case arriving aircraft has not been advised of any runway for landing, Runway 27 shall be considered as assigned runway for such arrival.

3.3 In case of departing aircraft from Lucknow returning on account of RCF, the departure runway of such aircraft shall be considered as assigned runway for landing.

3.4 Lucknow Aerodrome control Tower shall ensure approach and runway lights of the assigned runway in 'SWITCHED ON' position for RCF aircraft irrespective of day/night/visibility/weather conditions. Approach lights in 'SWITCHED OFF' position shall indicate that the approach should be made from the opposite end of runway. Runway and Approach lights on both sides in 'SWITCHED OFF' position shall indicate non- availability of Runway for aircraft experiencing RCF

3.5 In strong tail wind conditions for assigned Runway, aircraft will carry out a missed approach on assigned runway and after following complete missed approach procedure, will carry out any of the published instrument approach procedure for the opposite runway

**4. PROCEDURE FOR ARRIVALS:**

**4.1 ARRIVING AIRCRAFT CARRYING PROCEDURAL APPROACH**

4.1.1 Continue on ATS route, (if flying offset re-join ATS route as soon as practicable), maintaining/descending to cleared flight level and join 'LKN' VOR hold as published.

4.1.2 If higher descend to 3500 FT in 'LKN' hold. Start descend at ETA or EAT if it has been passed by the controller and acknowledged by the pilot of the RCF aircraft. Leave 'LKN' VOR at 3500 FT to carry out any of the published instrument approach procedure for assigned runway.

4.2 ARRIVING AIRCRAFT BEING VECTORED

4.2.1 Aircraft being vectored for approach, on experiencing RCF, shall maintain last assigned level and heading for 2 minutes following:

i. The time the last assigned level or FL 90 whichever higher is reached or

ii. the time the transponder is set to Code 7600 or the ADS-B transmitter is set to indicate the loss of air-ground communications; whichever is later and then take a convenient turn to proceed direct to 'LKN' VOR descending/maintaining last assigned level, or 3500 FT whichever is higher. If higher, descend to 3500 FT in 'LKN' hold.

4.2.2 Leave 'LKN' VOR at 3500 FT to carry out any of the published instrument approach procedure for assigned runway.

**5. PROCEDURE FOR DEPARTURES:**

**5.1 DEPARTURE INTENDING TO CONTINUE TO DESTINATION**

5.1.1 Departing aircraft experiencing RCF and intending to continue to its filed flight plan destination shall:

5.1.2 Continue on assigned heading climbing to or maintaining cleared level or FL 90, whichever is higher for 3 minutes following:

i. The time the last assigned level or FL 90 whichever higher is reached or

ii. the time the transponder is set to Code 7600 or the ADS-B transmitter is set to indicate the loss of air-ground communications; whichever is later and thereafter adjust level and speed in accordance with the filed flight plan; and proceed by the most direct manner possible to re-join the current flight plan route no later than the next significant point, maintaining cleared flight level/FL 90 whichever is higher.

5.2 DEPARTURE INTENDING TO LAND BACK AT LUCKNOW:

5.2.1 Departing aircraft experiencing RCF and intending to land back at Lucknow airport shall

i. Continue on Radar heading climbing to or maintaining the cleared Flight level or Flight level 90 whichever is higher.

ii. One minute after setting Mode A/C code 7600 or reaching the appropriate level as given in 5.2.1, whichever is later; take a convenient turn avoiding all restricted airspace as far as practicable and proceed direct to 'LKN' VOR to join hold as published.

iii. Descend to 3500 FT in 'LKN' hold. Leave 'LKN' VOR at 3500 FT to carry out any of the published instrument approach procedure for assigned runway.

**VILK AD 2.21 NOISE ABATEMENT PROCEDURES**

**NIL**

**VILK AD 2.22 FLIGHT PROCEDURES**

**I. LOW VISIBILITY PROCEDURES:**

**1.DEFINITIONS AND ABBREVIATIONS**

1.1 Instrument approach operations: An approach and landing using instruments for navigation guidance based on an instrument approach procedure. There are two methods for executing instrument approach operations:

a) a two-dimensional (2D) instrument approach operation, using lateral navigation guidance only; and

b) a three-dimensional (3D) instrument approach operation, using both lateral and vertical navigation guidance.

**Note**. - Lateral and vertical navigation guidance refers to the guidance provided either by:

a) a ground-based radio navigation aid; or

b) Computer-generated navigation data from ground-based, space-based, self-contained navigation aids or a combination of these

Instrument approach operations shall be classified based on the designed lowest operating minima below which an approach operation shall only be continued with the required visual reference as follows:

a)Type A: a minimum descent height or decision height at or above 75 m (250 ft); and

b)Type B: a decision height below 75 m (250 ft). Type B instrument approach operations are categorized as:

i.Category I (CAT I): a decision height not lower than 60 m (200 ft) and with either a visibility not less than 800 m or a runway visual range not less than 550 m.
ii.Category II (CAT II): a decision height lower than 60 m (200 ft) but not lower than 30 m (100 ft) and a runway visual range not less than 300 m; and
iii.Category III (CAT III): a decision height lower than 30 m (100 ft) or no decision height and a runway visual range less than 300 m or no runway visual range limitations.

1.2 Precision approach (PA) procedure: An instrument approach procedure based on navigation systems (ILS, MLS, GLS and SBAS Cat I) designed for 3D instrument approach operations Type A or B.

1.3 Decision Altitude/Height (DA/H): A specified altitude or height in a 3D instrument approach operation at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.

1.4 ILS Critical Area: An area of defined dimensions about the localizer and glide path antennas where vehicles, including aircraft, are excluded during all ILS operations. The critical area is protected because the presence of vehicles and/or aircraft inside its boundaries will cause unacceptable disturbance to the ILS signal-in-space.

1.5 ILS Sensitive Area: An area extending beyond the critical area where the parking and/or movement of vehicles, including aircraft, is controlled to prevent the possibility of unacceptable interference to the ILS signal during ILS operations. The sensitive area is protected to provide protection against interference caused by large moving objects outside the critical area but still normally within the airfield boundary.

1.6 Low Visibility Procedures: Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs.

1.7 Low Visibility Take-Off: A term used in relation to flight operations referring to a take-off on a runway where the RVR is less than 400 m.

1.8 Obstacle Free Zone (OFZ): The airspace above the inner approach surface, inner transitional surfaces, and balked landing surface and that portion of the strip bounded by these surfaces, which is not penetrated by any fixed obstacle other than a low-mass and frangible mounted one required for air navigation purposes.

1.9 Runway Visual Range (RVR): The range over which the pilot of an aircraft on the centerline of a runway can see the runway surface markings or the lights delineating the runway or identifying its centerline.

1.10 Safeguarding Procedures (SP): Safeguarding Procedures (SP) are instructions for relevant airport departments and airside operators to prepare ground services and facilities for low visibility operations, in order that when LVP are implemented all Safeguarding procedures are complete and airport is configured for CAT II/CAT III Operations and Low Visibility Take-Offs.

1.11 Aerodrome Operating Minima: The limits of usability of an aerodrome for -
a) take-off, expressed in terms of runway visual range and/or visibility and, if necessary, cloud conditions.
b) landing in 2D instrument approach operations, expressed in terms of visibility and/or runway visual range; minimum descent altitude/height (MDA/H) and, if necessary, cloud conditions; and
c) landing in 3D instrument approach operations, expressed in terms of visibility and/or runway visual range and decision altitude/height (DA/H) appropriate to the type and/or category of the operation.

1.12 Touchdown Zone- The portion of a runway, beyond the threshold, where it is intended landing aeroplane first contact the runway.

1.13 Visibility - Visibility for aeronautical purposes is the greater of -
a) The greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background.
b) The greatest distance at which lights in the vicinity of 1000 candelas can be seen and identified against an unlit background.

Note 1 - The two distances have different values in air of a given extinction coefficient, and the latter b) varies with the background illumination. The former a) is represented by the meteorological optical range (MOR).

Note 2 - The definition applies to the observations of visibility in local routine and special reports, to the observations of prevailing and minimum visibility reported in METAR and SPECI and to the observations of ground visibility.

1.14 Missed approach procedure- The procedure to be followed if the approach cannot be continued.

1.15 The abbreviations used in description to Low Visibility Procedures have the following meaning.

ARFF: Aerodrome Rescue & Fire Fighting
ATC: Air traffic Control
ATIS: Aerodrome Terminal Information Service
CCR: Constant Current Regulator
FAP: Final Approach Point
ILS: Instrument Landing System
LOC: Localizer
LSA: Localizer Sensitive Area
LVP: Low Visibility Procedure
MET: Meteorology
MID: Mid-Point
OFZ: Obstacle Free Zone
RVR: Runway Visual Range
SMC: Surface Movement Control
SP: Safeguarding Procedures
SSO: Technical/Communication Shift Supervisory Officer
TDZ: Touch-down Zone
WSO: Watch Supervisory Officer
ASMGCS: Advanced Surface Movement Guidance Control System

**2. INTRODUCTION**

**2.1 General:**

2.1.1 RWY 27 at Chaudhary Charan Singh Airport, Lucknow is equipped for the CAT II/CAT III Operations.

2.1.2 The following equipment shall be serviceable to the required standards to support CAT II/CAT III Operations:

a)ILS localizer, Glide Path and ILS DME
b)Aerodrome Ground Lighting System
c)Instrumented RVR system (TDZ, MID & END)
d)Standby power supply for ILS and aerodrome ground lighting system
e)ASMGCS for CAT II is desirable while for CATIII operations and LVTO (for RVR below 400 m), ASMGCS is mandatory.

2.1.3 It will be the responsibility of the Pilot to decide the category of ILS approach he/she may wish to carry out under the given conditions.

**2.2 Safeguarding Procedures (SP)**

2.2.1 Safeguarding Procedures are the mandatory actions to prepare airport for CAT II/CAT III operations (Low Visibility Procedures). They include inspection of aerodrome ground lighting, termination of all work in progress and removal of all equipment/ man material from localizer and glide path critical/ sensitive area and the manoeuvring area, restrictions on the movement of vehicles on the area.

2.2.2 WSO/Aerodrome Control Tower at Chaudhary Charan Singh Airport, Lucknow shall coordinate with Apron Control and subsequently the apron Control will coordinate with all the concerned agencies for implementation of Low Visibility Procedures.

2.2.3 SP shall be implemented whenever ATC considers the introduction of Low Visibility Procedures is necessary based on criteria in para 3.

**2.3 Low Visibility Procedures (LVP)**: Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs.

2.3.1 LVP shall only be implemented when Safeguarding Procedures (SP) have been completed, and the airport is configured for low visibility operations.

**2.4 ATC Requirement:**

2.4.1 WSO (ATC) shall implement and cancel LVP when so required based on criteria in para 3 and inform all concerned.

2.4.2 When any equipment listed in Para 2.1.2 above becomes unserviceable during periods of LVP, the concerned ATC unit shall advise the aircraft and accordingly CAT II/CATIII operations shall be suspended and information to this effect shall be included in ATIS broadcast.

2.4.3When Safeguarding Procedure (SP) is implemented ATC shall select the appropriate aerodrome ground lighting system CAT II/ CAT III facilities. These facilities shall remain selected until SP and LVP are cancelled

**2.5 ILS Critical Areas and Sensitive Areas**

2.5.1 The ILS critical and sensitive areas have been shown in the chart at Annexure-2.

Note: Signage indicating the limits of localizer and glide path sensitive areas is provided

2.5.2Chart indicating the critical and sensitive area of ILS is also available with apron control, ATC and WSO

**2.6 Reporting RVR**

2.6.1 There are three RVR transmissometer located at Touch-down Zone (TDZ), Mid-point (MID) and End Point Zone (END). The reference RVR value for the implementation and cancellation of LVP shall be the lower of the TDZ, MID & END RVR

2.6.2 When reporting RVR to pilots, the TDZ RVR shall always be intimated for the concerned Landing RWY.

**3. IMPLEMENTATION OF SAFEGUARDING PROCEDURES (SP) AND LOW VISIBILITY PROCEDRUES (LVP)**

**3.1 Criteria for Implementing Safeguarding Procedures (SP)**

3.1.1 Safeguarding Procedures shall be initiated when:

a) The visibility/ RVR is less than 1200 m or visibility/ RVR is forecast to deteriorate to 800 m or less; and/or

b) The cloud ceiling is 400 ft and forecast to fall to 200 ft or less.

**3.2 Criteria for Implementing Low Visibility Procedures (LVP)**

Low Visibility Procedures shall be implemented when -

a) Either, TDZ, MID or END RVR is less than 800m; and/or

b) Cloud ceiling is less than 200ft; and

c) Safeguarding Procedures (SP) have been completed and the airport safeguarded.

NOTE: Though LVP is implemented when RVR is less than 800 m, ILS CAT I operations will continue till TDZ RVR is not less than 550 m.

**3.3 Implementation of Safeguarding Procedures (SP):**

3.3.1 On the receipt of forecast for Low Visibility and/ or cloud ceiling, the Watch Supervisory Officer will inform SSO/ Technical Supervisor, Tower Supervisor, Approach Control Supervisor, Duty MET Officer and Apron control.

3.3.2 On receipt of the above information the above-mentioned agencies will take action for proper planning for the activation of LVP.

Note 1: Action to be taken by various agencies are described in SL. No. 7. When all the concerned agencies have completed their necessary actions, they shall report to Apron Control that their Safeguarding Procedure (SP) is completed, and the airport is safeguarded for LVP operations ,then Apron Control shall report to WSO/ATC that the Airport is safeguarded.

Note 2: Though LVP is implemented when RVR is less than 800 m, ILS CAT I operations will continue till TDZ RVR is not less than 550 m.

**3.4 Implementation of Low Visibility Procedures**

3.4.1 On receipt of confirmation of completion of all the required actions by the various agencies mentioned in Para 3.3.1, WSO/Aerodrome Control Tower shall implement Low Visibility Procedure when either TDZ, MID or END RVR is less than 800 m and/or the cloud ceiling is less than 200ft.

The implementation of LVP shall be intimated to (by WSO/Aerodrome control tower):

•

Approach Controller.

•Shift Supervisory Officer (SSO)

•Duty Met. Officer

•Apron Control

•Fire Station

3.4.2 WSO/ Aerodrome Control Tower will also ensure that “LOW VISIBILITY PROCEDURES IN FORCE” is included in ATIS Broadcast.

3.4.3 Aerodrome Tower Controller/WSO on being notified that LVP are to commence, will inform-

a)Fire station; and Apron Control

**3.5 Cancellation of Safeguarding Procedures (SP) & Low Visibility Procedures (LVP)**

3.5.1 WSO may terminate LVP when -

a) Meteorological conditions improve and TDZ, MID & END RVR are 800 m or more and/or the cloud ceiling is 200 ft or higher, and trend is for improvement for both runways.

b) Facilities, equipment and services necessary for CAT II/CAT III operations are degraded and/or the prevailing conditions are considered unsafe for such operations.

3.5.2 WSO should consult Meteorological Office for forecast before cancelling SP and LVP.

3.5.3 On cancelling LVP, Aerodrome Control Tower shall include it in the subsequent two ATIS broadcasts that “LOW VISIBILITY PROCEDURES ARE CANCELLED”. Aerodrome Tower Controller will inform all the agencies concerned as specified at Para 3.4.1.

3.5.4 If SP are implemented and LVP are not subsequently initiated and meteorological conditions improve and the visibility/ RVR is more than 1200 m and/or the cloud ceiling is 400 ft or higher and both are forecast to remain above the required SP criteria, WSO may cancel SP

## 4.LOW VISIBILITY PROCEDURE OPERATIONS

### 4.1 Approach Control Procedures

4.1.1 During LVP the approach controller and Tower controller shall have the following information:

•Status of ILS.

•Serviceability of visual aids.

•RVR information

•serviceability of ASMGCS

Note: Any degradation in any of the above facilities shall be immediately intimated to the arriving and departing aircraft by Approach/Tower controller.

4.1.2 In addition to the information normally transmitted by Approach control, following information shall be intimated to the arriving aircraft on first contact or as soon as possible thereafter- The current TDZ RVR, and

a) For CAT II operations - If TDZ RVR is below 550 m then MID RVR shall also be intimated. TDZ & MID RVR shall be available.

b) For CAT III operation - If TDZ RVR is below 300 m, then MID & END RVR readings shall also be intimated. TDZ, MID and END RVR shall be available.

c) The un-serviceability of any component parts of CAT II/CAT III facilities not previously broadcast on ATIS.

4.1.3. During CAT II/CAT III operations, Approach Control shall ensure that subsequent arrival is cleared for CAT II or CAT III ILS approach only after preceding arriving aircraft has landed and vacated the runway or has carried out missed approach. This is to ensure that the Critical and Sensitive area of ILS is not infringed.

Note: The departing aircraft must be airborne and cross the LOC antenna before the arriving aircraft has crossed the FAP. This is to ensure that the departing aircraft is clear of the ILS Sensitive and Critical Areas.

### 4.2 Aerodrome Control Procedures

4.2.1 Arriving aircraft should be issued landing clearance before it crosses the FAP.

4.2.2 Arriving aircraft should be given an unimpeded taxi route to allow it to clear the localizer sensitive area expeditiously.

4.2.3 Landing clearance shall not be issued until:

a) Preceding landing aircraft has vacated the runway and is clear of the localizer sensitive area (LSA), or preceding landing aircraft, carrying out missed approach, has passed over the localizer antenna.

b) Preceding departing aircraft is airborne and has passed over the localizer antenna

4.2.4 The ILS Critical & Sensitive Area in front of an arriving aircraft shall not be infringed from the time the arriving aircraft crosses the Final Approach Point (FAP) until it has completed its landing roll.

4.2.5 The Low Visibility Taxi Routes are intended to assist the pilots in determining their location on the airport during the periods of low visibility.

4.2.6 Aerodrome Control tower shall initiate immediate emergency action if an aircraft is not seen or not in radio contact as expected.

### 4.3 Surface movement Control Procedures:

4.3.1 Pilots need additional guidance and information when taxiing during periods of reduced visibility. The view from the cockpit of the aircraft is very limited. Therefore, taxi instructions and essential traffic information should be passed in a clear and concise manner.

4.3.2 Taxiing aircraft should be routed in accordance with the prescribed Low Visibility Routes to ensure a simple one-way traffic flow is maintained, unless otherwise necessitated due to operational requirements.

4.3.3 Surface Movement Control/Tower shall monitor the status of taxiway lights and immediately advise the aircraft under its control of any unserviceability affecting the LVP taxiways.

4.3.4 Surface Movement Control/Tower should monitor the progress of arriving aircraft as they vacate the runway after landing and ensure that they do not stop within the limits of ILS (LOC & GP) Critical & Sensitive Area, thereby degrading ILS integrity for subsequent landing aircraft.

4.3.5 Pilots shall report RWY vacated when aircraft is clear of the ILS sensitive area. Runway vacation boards have been provided at a distance of 160 m from the runway centerline on TWY A, B, C, D, E,P1and P3and P6.

4.3.6 When RVR is less than 800 m, vehicle movement should be restricted. Only operationally essential vehicles duly authorized by Surface Movement Control/Tower Controller should be permitted to operate. These vehicles shall remain outside the ILS (LOC & GP) Critical & Sensitive Area. Any movement of vehicle on the manoeuvring area shall be coordinated with ATC.

### 4.4 Low Visibility Procedure Taxi Route

4.4.1 When LVP is in force, aircraft shall be routed in accordance with the pre-designated taxi routes, unless otherwise necessitated due operational requirements.

4.4.2 During CAT II or CAT III ILS visibility conditions i.e. when RVR is reduced to less than 550m, ‘Follow me’ service will be provided to arriving/departing aircraft ‘on request’.

Note: Follow me service shall be provided by APRON CONTROL

4.4.3 The following taxi routes should be used for arrivals-

a) After landing on RWY 27 vacation via TWY D, E, P1 or P3 and then taxi to parking stands 101L to 14 as applicable

Note: Airline operators will be responsible for ensuring that the parking stand is clear of all equipment when aircraft is taxiing in for parking (Airlines shall deploy wing walkers with lighted marshaling wand for the same)

b) Taxi routing for departures - Taxi to CAT II/ III holding position RWY 27 via TWY 'B' or 'C' from stand No. 101L to 14:

Note 1: The airline operators shall ensure that push-back area is clear of all equipment before push back is commenced.

Note 2: In order to maintain a unidirectional taxi routing during LVP; the AOCC/Apron Control may minimize allocation of Stand No.(s) 103L, 103 and 103R in case the aircraft to be parked on these bays are required to be pushed back/start during the LVP operations

4.4.4 The following taxi routes shall be used for arrivals/ departures from stand 212 to stand 216.

i) After landing on RWY 09, vacate via TWY P6, follow TWY P - TWY L3 – TXL L1 to parking stands 212 to 216 as applicable. For departure, taxi out via TWY L3-TWY P -TWY P6 vacate via TWY B, C, D, E or P3 then re-enter the RWY via TWY P1.

ii) After landing on RWY 27, vacate via TWY C, D, E, P1 or P3, then re-enter the runway via any TWY and vacate again via TWY P6 – TWY P – TWY L3-TXL L1. For departure, taxi out via TWY L3- TWY P -TWY P6.

**5.DESCRIPTION OF EQUIPMENT**

**5.1 Runway Visual Range (RVR)**

5.1.1 There are three transmissometer recording RVR values for RWY 27 one unit is located at the touch down zone (TDZ) and others unit at runway mid-point (MID) and endpoint (END). RVR values always refer to as Touchdown RVR (TDZ) and Mid-point RVR (MID) and end-point (END).

5.1.2 RVR is reported in the following scales:

a) In the increments 25 m when less than 400 m.

b) In the increments 50 m when RVR greater than 400 m but less than 800 m.

c) In the increments 100 m when greater than 800 m.

d) RVR is reported throughout periods when either horizontal visibility and /or the runway visual range is less than 1500m. The maximum reportable value of RVR is 2000 m. When the horizontal visibility is 2000m or less, but RVR is more than 2000 m, it is reported as RVR ABOVE 2000 M

5.1.3 RVR serviceability for CAT II/ CAT III operations:

a) For CAT II operations TDZ and MID RVR shall be available.

b) For CAT III operations TDZ, MID and END RVR shall be available.

**5.2 Aerodrome Ground Lighting System**

5.2.1 The Precision Approach lighting system for CAT II/CAT III operations are installed on RWY 27 at Chaudhary Charan Singh Airport, Lucknow.

5.2.2 During CAT II/ CAT III operations, the generator will take over as primary power source and the mains supply becomes the backup power source.

5.2.3 The taxiway B, C, D, E, P1, P3 and P6 have CAT III standard taxiway lighting.

5.2.4 Parking stands from 101L to 14 and stand 212 to 216 have been provided with CAT III Centerline lighting system.

5.2.5 STOP BAR: Stop bars have been provided on the following TWY: A B,C, D, E, P1 and P3.and P6

5.2.6 When LVP is in force the aerodrome ground lighting must comply with the following minimum serviceability requirement

| Aerodrome Ground Lighting Facility | CATII/CATIII Unserviceability | Restrictions |
|---|---|---|
| **Approach Lights** | The inner 450 meters- more than 5 % of all lights | Suspend CAT II/CAT III operations |
| | Beyond 450 meters more than 15% of all lights | |
| **Runway Edge lights** | More than 5% of all Lights | Suspend CAT II/CAT III operations |
| | Two adjacent lamps | |
| **Runway centre line lights** | More than 5% of all Lights | Suspend CAT II/CAT III operations |
| | Two adjacent lamps | |
| **Touchdown Zone lights** | More than 10% of all Lights | Suspend CAT II/CAT III operations |
| | Two adjacent lamps | |

| Aerodrome Ground Lighting Facility | CATII/CATIII Unserviceability | Restrictions |
|---|---|---|
| **Threshold lights** | More than 5% of all Lights | Suspend CAT II/CAT III operations |
| | Two adjacent lamps | |
| **Runway End lights** | More than 25% of all Lights | Suspend CAT II/CAT III operations |
| | Two adjacent lamps | |
| **Taxiway centre line lights** | Not applicable to CATII operation | Close affected taxiways, use alternate taxi route. |
| | Two adjacent lamps | |
| **Standby Generators** | Generators in any one substation | Suspend CAT II/CAT III operations in the affected runway |

*Note: An unserviceability of any of the following facilities does not affect the CAT II/CAT III operations:*
*a) PAPI*
*b) Taxiway edge lights on curves*

**5.3 RVR/Visibility for Take-off (Commercial Transport Aeroplanes)**

| Take-off RVR/Visibility | |
|---|---|
| Facilities | RVR/ VIS (Note 1)<br>Cat A, B, C & D |
| Adequate Visual reference.<br>Refer Note 2 below (Day only) | 500 m |
| Runway edge lights or Runway centre line markings 3<br>Refer Note 3 below | 400 m |
| Runway edge lights and Runway centre line markings 3<br>Refer Note 3 below | 300 m |
| Runway edge lights and Runway centre line lights | 200 m |
| Runway edge lights and Runway centre line lights and relevant RVR information refer 4 note 4 | 150 m |
| High intensity Runway edge lights and Runway centre line lights (spacing 15 m or less) and relevant RVR information refer 4 note 4 | 125 m |
| High intensity Runway edge lights and Runway centre line lights (spacing 15 m or less), approved lateral guidance system and relevant RVR information refer 4 note | 75 m |
| | |

Note 1 – The TDZ RVR/VIS may be assessed by the pilot.
Note 2 - Adequate Visual reference means that a pilot is able to continuously identify the take-off surface and maintain directional control.
Note 3 - For night operations at least runway edge lights or centre line lights and runway end lights are available.
Note 4 - The required RVR must be achieved for all relevant RVR reporting points (touchdown, mid- point and stop-end/roll-out). The governing RVR shall be the lowest of the reported RVRs.
Note 5 – Aircraft need to lineup abeam TWY E for RWY 09 during LVTO (As the RVR instrument is located near TWY-E).

**5.4 Inspections of Aerodrome Ground Lighting System**

5.4.1 One of the LVP criteria is that the appropriate aerodrome ground lights must be inspected during the hour preceding implementation of LVP, and thereafter every subsequent two-hour period. The lighting inspections should be accorded with high priority and for this purpose aircraft operations may have to be delayed if necessary.

5.4.2 Apron control is responsible for organizing lighting inspections when WSO/Aerodrome Control Tower informs that Safeguarding Procedures are to be implemented. He shall arrange an inspection of the relevant aerodrome ground lighting system. To ensure minimum delay in completing the inspection, separate teams may inspect the runway in use, associated taxiways and apron area.

5.4.3 For SP and LVP only the lighting for the active runway and associated taxiways are inspected.

**5.5 Non-Visual Ground Surveillance System:**

5.5.1 Chaudhary Charan Singh Airport, Lucknow is equipped with Advanced Surface Movement Guidance Control System. The system provides non-visual electronics surveillance of manoeuvring areas to facilitate the controllers to identify potential ground conflict and runway incursions.

5.5.2 For CAT III Operations, availability of Non-Visual Surveillance System such as Advanced Surface Movement Guidance Control System is mandatory

## 5.6 Navigational Aids

5.6.1 RWY 27 has been equipped with an Instrument Landing System (ILS) for CAT I. CATII/ CAT III.

5.6.2 The ILS Category Monitor Panel at the Control Tower console indicates the ILS category availability by monitoring the following equipment:

a. Main and standby localizer transmitters

b. Main and standby glide path transmitters

5.6.3 The status of the ILS and DME is monitored and displayed by a separate nav-aid status indicator panel.

5.6.4 ILS equipment serviceability required for CAT II/CAT III operations: -

a. Both main and standby localizer transmitters.

b. Both main and standby glide path transmitters.

c. Stand by power supply for each unit.

d. ILS DME

**5.7 Airport Rescue and Fire Fighting Services (ARFF)**

5.7.1 The ARFF shall be on Weather Standby Position whenever LVP is in force. Following predetermined positions will be taken by ARFF vehicles when LVP are in force:

a) Near Glide Path

b) In front of Fire station near the approach road to runway

5.7.2 In the event of an incident when LVP are in force, Aerodrome Control / SMC should provide the maximum assistance in directing ARFF to the required location.

**6.SUMMARY OF THE LOW VISIBILITY PROCEDURES**

**6.1 Vehicular movement**

6.1.1 Vehicles shall not be cleared to enter/ cross the runway once an inbound aircraft has crossed the Final Approach Point (FAP).

6.1.2 Vehicular movement on the manoeuvring area shall be restricted to essential vehicles only. The Record of all such vehicles shall be maintained by Apron control and shall be communicated to ATC.

6.1.3 Vehicles shall not be held at any point closer to the runway than the CAT II/CATIII holding position/ stop bars.

**6.2 Aircraft movement**

6.2.1 Aircraft shall not be held at any point closer to the runway than the CAT II/CAT III holding position/stop-bars.

### 6.3 ATC Procedures:

6.3.1 Departing aircraft must cross the LOC antenna before the arriving aircraft has crossed the Final Approach Point (FAP).

6.3.2 The aim will be to give landing clearance to the arriving aircraft before it crosses the Final Approach Point (FAP).

6.3.3 Protection of localizer/glide path and critical and sensitive area must be ensured.

6.3.4 The ILS Critical & Sensitive Area in front of a departing aircraft shall not be infringed from the time take-off clearance is issued until the aircraft has departed and passed over the localizer aerial.

6.3.5 During CAT II/CAT III operations, Approach Control shall ensure that subsequent arrival is cleared for CAT II or CAT III ILS approach only after preceding arriving aircraft has landed and vacated the runway or has carried out missed approach. This is to ensure that the Critical and Sensitive areas of ILS are not infringed

**7.ACTIONS TO BE TAKEN BY VARIOUS AGENCIES**

7.1 Before commencement of winter season, a meeting will be held by CAO, LIAL Lucknow in the month of October/ November every year to

inform all airlines and agencies operating at airport about their roles/ responsibilities and create awareness to ensure cooperation for safe

airport operations during periods of low visibility.

7.2 All the agencies shall ensure that staff and drivers are suitably trained during CAT II/CAT III operations.

7.3 A refresher program for ATCO's and personnel responsible for airside operations shall be conducted every year before LVP are expected
to be in place.
7.4Action by Watch Supervisory Officer (WSO)/Aerodrome control Tower:
7.4.1 Implementing Safeguarding Procedures
7.4.2 Implementing LVP
7.4.3 For the purpose of commencing Low Visibility Procedure, WSO/ Aerodrome Control Tower shall inform:
•SSO/Duty Officer (Equipment Room)
•Duty Met. Officer
•Apron Control
7.4.4 WSO/Aerodrome control tower would declare LVP effective after confirmation from concerned wings, all actions have been completed.
7.4.5 WSO/ Aerodrome Control Tower shall also ensure that "LOW VISIBILITY PROCEDURES IN FORCE" is included in ATIS broadcast.
7.4.6 WSO/Tower Supervisor may terminate LVP in consultation with Duty Met. Officer when:
7.4.6.1 Prevailing meteorological conditions improve and both the TDZ RVR and MID RVR are 800 m or more; and
7.4.6.2 Cloud ceiling is 200 ft or higher, and trend is for improvement. OR
7.4.6.3 Facilities, equipment and services necessary for CAT II/CAT III operations are degraded and/or the prevailing conditions are considered unsafe for such operations.
7.4.7 WSO/ Aerodrome Control Tower will intimate SSO/Duty Officer (Equipment Room) regarding the termination of LVP operation

**7.5 Action by Tower Controller and Approach Controller:**
7.5.1 On being notified by WSO/ Aerodrome Control Tower that ILS CAT II/CAT III Low Visibility Procedures are to commence, the Aerodrome Tower Controller will:
•Inform Aerodrome Rescue & Fire Fighting Services for Weather Standby Position
•Check ILS status.
•Check aerodrome ground lighting is correctly selected and operating properly.
•Check transmissometer display.
•Check status of ASMGCS

7.5.2 After the commencement of ILS CAT II/CAT III operations, the Aerodrome Tower Controller will Inform the arriving aircraft "ILS CAT II/CAT III Low Visibility Procedures in operation"
7.5.3 Inform the arriving aircraft "ILS CAT II/CAT III Low Visibility Procedures in operation".
Note: After an aircraft is 8 NM from Touch down RVR observations need not be passed unless there is a change in RVR values.

7.5.4 Provide adequate spacing between the successive arriving aircraft so that landing clearance may be issued to the preceding aircraft before it crosses the Final Approach Point (FAP). Consideration should also be given for a departing aircraft using full length of runway for take-off.

7.5.5 Consideration should be given for the time taken by the previous landing aircraft to clear the critical area of the localizer. Give an unimpeded taxi route to arriving aircraft to allow it to clear the localizer sensitive area expeditiously.

7.5.6 Inform pilots/ approach controller of failures of ILS, lighting system, transmissometer relevant to ILS CAT II/CAT III Low Visibility Operations.

7.5.7 Initiate emergency action if aircraft on CAT II/ CAT III ILS is not seen or is not in radio contact as expected.

7.5.8 Record of the above actions with time be maintained and signed by the officer taking action.

a)During the period the Low Visibility Procedures are effective:
b)Monitor surface movement of aircraft and essential vehicles in the manoeuvring area.
c)Inform all taxiing aircraft of the preceding taxiing or holding aircraft

**7.6 Action by SSO**
7.6.1 On receipt of 'Forecast for LVPs' from the WSO/Tower Supervisor, SSO will ensure the –
7.6.1.1 Status of:
7.6.1.1.1 Serviceability of Main and standby ILS system (LOC/Glide Path/ILS-DME); and
7.6.1.1.2 Indication of nav- aids facilities in ATC units.
7.6.1.1.3 Serviceability of ASMGCS

7.6.1.2 Inform WSO/Tower Supervisor of any unserviceability in the equipment which is likely to affect ILS CAT II/CAT III operation.
7.6.1.3 On receipt of 'Advisory Message' from WSO/Aerodrome Control Tower that LVPs are to be made effective, SSO will intimate WSO/ Aerodrome Control Tower in case of any unserviceability which may affect ILS CAT II/CAT III operations.

**7.7 Apron Control**

7.7.1 After initiation of Safeguarding Procedures, Apron Control shall ensure that no unauthorized vehicle/person enters or present in the sensitive/critical areas of localizer and glide path.

7.7.2 All civil/electrical works-in-progress to be stopped immediately and men/material/equipment to be removed from the sensitive/ critical areas of localizer glide path and OFZ.

7.7.3 After ensuring above, Apron Control will confirm the same to the WSO/Tower Supervisor.

7.7.4 Ensure that no workers/vehicles (including grass cutting contractor, garbage removal contractor, electrical/civil contractor, etc.) shall enter the operational area during the operations of ILS CAT II/CAT III Low Visibility Operations are in progress

**7.8Action by Shift manager (Electrical):**

On receipt of information to implement Low Visibility Procedures from apron control, the Shift Manager Electrical Section will:

7.8.1 Check in following aerodrome ground lighting associated with RWY 27 is serviceable and can be operated at full intensity.

•Approach lighting system.
•Runway Edge lights
•Runway threshold and end lights
•Runway centerline lights
•Runway touch-down zone lights
•Stop Bar lights
•Taxiway edge lights
•Taxiway centerline Lights.
•Runway holding position lights
•Runway guard Lights

Note: No adjustment of light intensities shall be made without permission from the Control Tower.

7.8.2 Inform the serviceability status of the above visual lighting aids to WSO/Aerodrome Control Tower.

7.8.3 Ensure that generator takes over as primary power source, and the mains supply becomes the backup (Secondary) power source.

7.8.4 Ensure that Constant Current Regulator Room (CCR) is manned by the electrical engineer or electrical technician for standby power supply requirements and shall remain available on telephone or maintain a listening watch on Walkie-Talkie.

7.8.5 Coordinate for regular inspection of aerodrome ground lighting and inform the unserviceability or any change in status of any facility/systems to WSO/Aerodrome Control Tower immediately

**7.9 Action by CISF and LIAL Security**

7.9.1 The inspector in charge shall ensure that movement of security personnel is restricted only through Gate No.2 and Gate 3.
7.9.2 Apron control shall coordinate with CISF to stop vehicular movement on the perimeter road around Runway 27 approach except emergency vehicles fitted with R/T and VELO and operating with prior clearance from ATC.
7.9.2.1 APRON CONTROL shall coordinate with CISF to close barrier on perimeter road near glide path and Bay No.101L. for stopping the vehicular movement except Follow Me Jeep, ARFF vehicles and AAI vehicles fitted with RT and VELO. Prior permission of ATC and Apron Control is essential to permit other vehicle on this perimeter road.
Note: Both the barriers shall be in a lower position whenever LVP is in progress.

**7.10Action by Fire Station**

7.10.1Ensure that ARFF are on Weather Standby Position at following pre-determined position (PDP) whenever LVP is in force.
•Near Glide Path
•In front of the Fire station near the approach road to runway

**7.11Action by CISF Inspector-In-Charge/CISF Control Room**
7.11.1The Inspector-In-Charge, on receipt of advice to implement Low Visibility Procedures, will immediately inform all access gates and CISF posts under their respective controls in operational area to restrict essential vehicles to use service road towards RWY 27 side only, for proceeding to localizer and vice-versa, till such time the termination of LVP.
7.11.2For carrying out Security Checks, Inspector In-Charge of CISF Control Room would be escorted by 'Follow-Me Jeep' for which they would liaise with APRON CONTROL.
7.11.3 The inspector in charge shall ensure that the movement of security personnel is restricted only through Gate No.2.

**7.12 Action by Duty Officer-Meteorology**
7.12.1 Duty Met. Officer would issue a 'Forecast for Low Visibility Procedures' to the Watch Supervisory Officer WSO/Tower Supervisor whenever he expects that the RVR RWY 27 and/or cloud ceiling will fall below 800 m and/or 200 ft or less respectively.
7.12.2 Whenever Duty Met. Officer anticipates that RVR RWY 27 is likely to fall below 800M and/or cloud ceiling to 200 ft or less within next 2 hours, he will issue an 'Advisory Message' to WSO/Tower Supervisor to this effect.
7.12.3 Whenever the RVR and/or cloud ceiling are 800M and/or 200 ft respectively and the trend is towards improvement in these elements of weather conditions, the Duty Met Officer may, when requested by WSO/Tower Supervisor, advise him about such improving weather conditions for the purpose of termination of LVP operation.
7.12.4 The Duty Met. Officer would ensure that the RVR displays in ATC units in the Control Tower and Approach Control are serviceable. He would also ensure that RVR/visibility recorders of Touch-down zone (TDZ), Mid-Point (MID) and END RVR positions are serviceable.
NOTE: Due to high variability of meteorological elements in space and time and the limitations of forecasting techniques available, it may not be always possible to issue a precise forecast of RVR particularly in case of transient phenomenon within two hours.
**7.13 Action by Other Agencies (Airlines, Refueling Companies, Catering Agencies, Customs, Immigration, Health etc.)**

7.13.1 All agencies operating in the operational area shall ensure that only those vehicles that are absolutely essential for aircraft operations in the operational area. The drivers of these vehicles should keep a look out for taxing aircraft and other vehicles to prevent incident/accidents.
7.13.2 All the vehicles must have their VELO/obstruction lights "ON" during Low Visibility Procedures operations

7.13.3Follow all instructions/sign boards provided for vehicular movement area/service roads.
7.13.4No vehicle/equipment/personnel shall enter in and around the vicinity of the runways or taxi-tracks except with prior permission of Aerodrome Control Tower.

**8.Termination of Low Visibility Procedures**

8.1When metrological conditions have improved and both the TDZ and MID/RVR are 800M or more, cloud ceiling is 200 ft or higher and trend is for improvement, WSO/Aerodrome Control Tower would terminate operations of LVPs. He may obtain advice from Duty Met. Officer regarding improvement in weather conditions for the purpose of termination of LVP operations.
8.2The WSO/ Aerodrome Control Tower will intimate SSO/Duty Officer (Equipment Room) regarding the termination of LVP operations.
8.3Aerodrome Tower Controller will in turn to inform all concerned/all the previously notified personnel to resume normal operations.

**9.Procedures for Protection of Critical Area and Sensitive Area of ILS during Low Visibility Operations**
**9.1Introduction**

9.1.1The occurrence of interference to ILS signals is dependent on the total environment around the ILS antennas, and the antenna characteristics. Any large reflecting objects, including vehicles or fixed objects such as structures within the radiated signal coverage, will potentially cause multipath interference to the ILS course and path structure. The location and size of the reflecting fixed objects and structures in conjunction with the directional qualities of the antennas will determine the static course or path structure quality whether Category I, II or III. Movable objects can degrade this structure to the extent that it becomes unacceptable. The areas within which this degradable interference is possible need to be defined and recognized. For the purposes of developing protective zoning criteria, these areas can be divided into two types, i.e. critical areas and sensitive areas:

a)The ILS critical area is an area of defined dimensions about the localizer and glide path antennas where vehicles, including aircraft, are excluded during all ILS operations. The critical area is protected because the presence of vehicles and/or aircraft inside its boundaries will cause unacceptable disturbance to the ILS signal-in-space
b)The ILS sensitive area is an area extending beyond the critical area where the parking and/or movement of vehicles, including aircraft, is controlled to prevent the possibility of unacceptable interference to the ILS signal during ILS operations. The sensitive area is protected against interference caused by large moving objects outside the critical area but still normally within the airfield boundary

**9.2Purpose**
9.2.1The purpose of these procedures is to issue guidelines to ATCOs and officials of other agencies working at Lucknow Airport so as to ensure that the critical and sensitive areas of ILS are protected during low visibility operations as per the guidelines given below:

**9.3Protection of ILS Critical & Sensitive Areas:**
9.3.1ATCOs shall comply with the following guidelines for protection of Localizer Critical/Sensitive area during CAT II/III Operations:
9.3.2Approach Control shall ensure that a succeeding arriving aircraft is cleared for CAT II or CAT III ILS approach in such a way that:
a)Preceding arriving aircraft has landed and vacated the runway and is clear of localizer critical and sensitive Area or has carried out the missed approach procedure and has passed over localizer antenna.
b)Preceding departing aircraft is airborne and has passed over the localizer antenna.
9.3.3Arriving aircraft should be given un-impeded taxi route to allow it to clear the localizer and Glide Path Critical and Sensitive area expeditiously.
9.3.4The Localizer and Glide Path Critical and Sensitive area in front of an arriving aircraft shall not be infringed from the time the arriving aircraft crosses the Final Approach Point until it has completed its landing roll.
9.3.5The ATCOs shall activate the ILS Critical and Sensitive area map saved in the ASMGCS.
9.3.6On receipt of advisory message from WSO/ Aerodrome Control Tower that LVP is to be made effective, CISF Staff (deployed round-the-clock in huts near the boundary wall near Glide Path Sensitive area) shall ensure that no unauthorized movement takes place in the portion of Glide path & LLZ sensitive area

## II.ATS Surveillance Services:

ADS-B/PSR/MSSR based ATS Surveillance services are provided to suitably equipped aircraft BTN 0030-1630 UTC within the coverage area of Lucknow Approach Control.
ATS surveillance based services using ASR/MSSR/ADS-B provided during the operations within the jurisdiction of Lucknow Area Control to suitably equipped aircraft within the coverage area of Lucknow ASR/MSSR/ ADS- B ground station on VHF 120.45 MHZ and standby 120.2 MHZ.
SSR Transponder/ADS-B equipped aircraft flying within Lucknow Area Control jurisdiction shall comply with relevant provisions.

## VILK AD 2.23 ADDITIONAL INFORMATION

1. Two 90FT high steel lattice Mast raised at Tx station located 264657.90N 0805301.70E lighted and 264656.90N 0805303.70E unlighted.

2. One lattice Mast for VHF link erected on top of Terminal Building with coordinates 264500.90N 0805253.70E Hgt 14.6M AGL Hgt of mast 136.6M AMSL. Mast is lighted.

3. Both lighted and unlighted landing Direction indicator (LDI) available at VILK. Position between TWY ‘B’ and TWY ‘C’

4. Pilot to exercise caution while Taxing IN/OUT with low power from Apron.

5.PERM commissioning of ASR-MSSR

6. ASMGCS commissioned to enhance visual surveillance.

7.Isolation bay suitable upto A330-200 AVBL at VILK.

8.Connecting TWY 'M' PCR: 2037/R/D/W/T WIDTH: 23M (MINIMUM).

9.Dimensions of Isolated Aircraft Parking Position: 208.6MX112.0M, PCR: 2037/R/D/W/T.

10. Take-off run available from different entry taxiways

11. Men and machinery deployed for grass cutting and bird scaring on Runway Strip at distance of 90M and beyond on either side of Runway Centerline. Height of machinery restricted to 3M AGL. Pilot to excercise caution while landing/take-off.

12.Intermediate Turn Pad RWY 27 Capable of Handling Code C Type of aircraft in CAT I Visibility Condition. TORA is 2214m for Intermediate Turn Pad RWY 27. PCR for Extended Portion of Intermediate Turn Pad RWY 27: 862/F/A/W/T.

| RUNWAY 27 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| ENTRY TAXIWAY | Intermediate Turnpad | A | B | C | D | E | M | P3 | P6 |
| TORA(M) | 2214 | 1187 | 907 | 784 | 560 | 317 | 317 | 107 | 1762 |
| RUNWAY 09 | | | | | | | | | |
| ENTRY TAXIWAY | | A | B | C | D | E | M | P3 | P6 |
| TORA(M) | | 1557 | 1837 | 1959 | 2183 | 2427 | 2427 | 2637 | 1019 |

## 13. Details of Taxiways Width, Surface and Strength etc.

| Designation | Taxiway/ Apron Width (M) | PCR | Taxiway/ Apron Length (M) | Surface | Remarks/ Suitability |
|---|---|---|---|---|---|
| TWY A | 23 | 943/F/B/W/T | 252 | Asphalt | Up to Code C aircraft |
| TWY B | 23 | 862/F/A/W/T<br>1175/R/A/W/T | 207 | Asphalt<br>Concrete | Up to Code E aircraft |
| TWY C | 23 | 862/F/A/W/T<br>1765/R/C/W/T | 207 | Asphalt<br>Concrete | Up to Code E aircraft |
| TWY D | 23 | 1506/R/B/W/T | 245 | Concrete | Up to Code E aircraft |
| TWY E | 23 | 1506/R/B/W/T | 223 | Concrete | Up to Code E aircraft |
| TWY M | 23 | 2037/R/D/W/T | 161 | Concrete | Up to Code E aircraft |
| TXL L | 23 | 862/F/A/W/T<br>1765/R/C/W/T | 960 | Bitumen Concrete | Up to Code E aircraft |
| TWY P1 | 23 | 1181/F/C/W/T | 207 | Bitumen | Up to Code E Aircraft |
| TWY P3 | 23 | 1181/F/C/W/T | 207 | Bitumen | Up to Code E Aircraft |
| TWY L4 | 23 | 862/F/A/W/T<br>1765/R/C/W/T | 110 | Asphalt/ Concrete | Up to Code C Aircraft |
| TWY P6 | 26.3 | 820/F/C/W/T | 150.5 | Asphalt | Up to Code E aircraft |
| TWY P | 23 | 830/F/C/W/T | 350 | Asphalt | Up to Code E aircraft |
| TWY L3 | 43.6 | 960/F/C/W/T | 70 | Asphalt | Up to Code E aircraft |
| TXL L1 | 23 | 1520/R/C/W/T | 260 | Concrete | Up to Code E aircraft |
| APRON 1 | | 1765/R/C/W/T | | Concrete | Up to Code C & E aircraft |
| APRON G | | 1765/R/C/W/T | | Concrete | Up to Code B aircraft |
| TURN APD 27 | -- | 943/F/B/W/T | -- | Bitumen | Up to Code E aircraft |
| INTERMEDIATE TURN PAD | -- | 862/F/A/W/T | -- | Bitumen | Up to Code C aircraft |

| East Apron | 128.5 | 580/R/C/W/T | 210.8 | Concrete | Up to Code C aircraft |
|---|---|---|---|---|---|

14. Details of Stands of Apron 1:

| Stand No. | Coordinates | Suitability | PCR | Elevation (FT) | Aircraft Type | Remarks | Contact/ Remote |
|---|---|---|---|---|---|---|---|
| 101L | 264548.787N 0805233.775E | A321/ B739 | 1460/R/C/W/T | 403 | Code C | Power-in/ Pushback | Remote |
| 101 | 264549.508N 0805234.533E | A359/ B773 | 1460/R/C/W/T | 403 | Code D/E | Power-in/ Pushback | Remote |
| 101R | 264549.544N 0805235.228E | A321/ B739 | 1460/R/C/W/T | 403 | Code C | Power-in/ Pushback | Remote |
| 102L | 264548.826N 0805236.707E | A321/ B739 | 1460/R/C/W/T | 404 | Code C | Power-in/ Pushback | Remote |
| 102 | 264549.549N 0805237.528E | A359/ B773 | 1460/R/C/W/T | 404 | Code D/E | Power-in/ Pushback | Remote |
| 102R | 264549.586N 0805238.162E | A321/ B739 | 1460/R/C/W/T | 404 | Code C | Power-in/ Pushback | Remote |

Note: All stands have been provided with aircraft stand identification signs and AVDGS

15.Details of Stands of New East Apron:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 212 | 264549.02121N 0805320.23929E | A321/ B739 | 580/R/C/W/T | 402.45 | Code A/ B/C | Power-in/ Pushback | Remote |
| 213 | 264549.04167N 0805321.74018E | A321/ B739 | 580/R/C/W/T | 402.55 | Code A/ B/C | Power-in/ Pushback | Remote |
| 214 | 264549.06413N 0805323.24278E | A321/ B739 | 580/R/C/W/T | 402.39 | Code A/ B/C | Power-in/ Pushback | Remote |
| 215 | 264549.08249N 0805324.74366E | A321/ B739 | 580/R/C/W/T | 402.29 | Code A/ B/C | Power-in/ Pushback | Remote |
| 216 | 264549.10371N 0805326.24649E | A321/ B739 | 580/R/C/W/T | 402.32 | Code A/ B/C | Power-in/ Pushback | Remote |

**Note:**
i) All stands have been provided with aircraft stand identification signs and AVDGS.
ii) Entry & Exit of New east apron via TWY P6 only.

**16**.Tug release point (TRP) available on apron taxilane to accommodate more pushback.
TRP 01 abeam stand 10.
TRP 02 abeam stand 07.
TRP 03 abeam on TWY L west of stand 103L.
ATC may give pushback and startup on TRP as per traffic conditions.

**VILK AD 2.24 CHARTS RELATED TO AN AERODROME**

1. Aerodrome Chart (09 July 2026)

2. Aircraft Parking/Docking Chart Apron-1 (09 July 2026)
3. Aircraft Parking/Docking Chart Apron-East (06 AUG 2026)
4. Aircraft Parking/Docking Chart Apron-G (17 April 2025)
5. Aircraft Parking/Docking Chart Isolation Bay (17 April 2025)
6. Aerodrome Obstacle Chart Type – A (Operating Limitations) RWY 09
7. Aerodrome Obstacle Chart Type – A (Operating Limitations) RWY 27
8. Take Off Run Available from Intersection RWY 09 (9 JULY 2026)
9. Take Off Run Available from Intersection RWY 27 (9 JULY 2026)
10. Precision Approach Terrain Chart RWY 27
11. Instrument Approach Chart - ILS CAT I Procedure RWY27(27 November 2025)
12. Instrument Approach Chart - ILS CAT II/IIIA/IIIB Procedure RWY27(27 November 2025)
13. Instrument Approach Chart - VOR Procedure RWY 09
14. Instrument Approach Chart - VOR Procedure RWY 27
15. ATC Surveillance Minimum Altitude Chart
16. Instrument Approach Chart - RNP Y Procedure RWY 27
17. Coding Table - RNP Y Procedure RWY 27
18. Instrument Approach Chart - RNP Y Procedure RWY 09
19. Coding Table - RNP Y Procedure RWY 09

## VILK AD 2.25 VISUAL SEGMENT SURFACE (VSS) PENETRATION

| Procedure | Procedure Minima | VSS Penetration |
|---|---|---|
| 1 | 2 | 3 |
| | | |